

LED Emergency Light Cum Power Bank Plus Mini Fan Driver

Gautam Kumar Mandal

This simple circuit can be used as an LED emergency light, a power bank for charging mobile phones, and as on-the-go (OTG) USB DC fan driver.

The power bank with 12Ah can deliver 2A current for charging a mobile phone. There is a provision for adjusting the charging rate through boost converter module. Also, there is a digital voltmeter that monitors Li-ion rechargeable battery pack's voltage as well as the

mobile charging voltage. Inside view of the author's prototype is shown in Fig. 1.

Following components would be required to build the project:

- Four 3,000mAh Li-ion 18650 type batteries (see Fig. 3)
- A 5V, 1A TP4056 micro USB battery charger module with inbuilt LED indicator for charging and automatic overcharge protection system
- A 4V white LED strip with aluminium-sheet base
- An XL6009 DC-DC boost converter module
- Two SPST switches
- One SPDT switch for the digital voltmeter
- A tiny 0.71cm (0.28-inch), 2.5V to 30V DC range mini digital voltmeter
- Switchboard's plastic enclosure with USB female sockets for mobile charging and OTG fan
- Transparent acrylic sheet for the emergency light on the front side
- An OTG mini fan

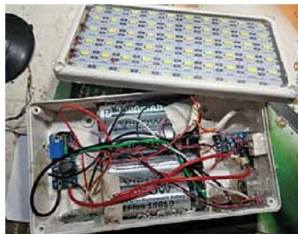


Fig. 1: Author's prototype with inside view

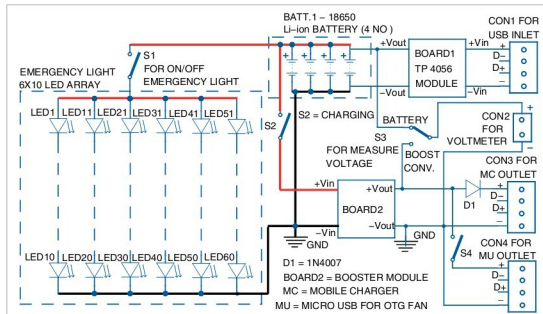


Fig. 2: Circuit diagram of the emergency light with power bank and OTG fan driver



Fig. 3: Li-ion battery pack

Circuit and working

The circuit diagram of the emergency light with power bank and OTG fan driver is shown in Fig. 2. It is built around TP4056 charging module (Board1), DC-DC booster module (Board2), four 18650 type Li-ion batteries (BATT.1), 6 x 10 LED-array, and a few other components.

TP4056 charging module used in the project is shown in Fig. 4 and XL6009 DC-DC booster module in Fig. 5.

Battery bank charger

The battery bank includes four Li-ion 18650 type cells that are connected in parallel. For charging the Li-ion battery bank, a TP4056 charging board (5V, 1A) with an LED indicator and automatic protection is used. Only one charging module is used in this project.

With a good quality smartphone charger input at CON1, it takes around two hours to fully charge the battery bank. When it is fully charged (battery voltage is 4.1 volts), the inbuilt red LED of TP4056 module becomes green. Though maximum voltage